

Combinatorial Maximalism

Exhaustion as Proof in Context-Bounded Software Construction:
the Algebra, Geometry, and Calculus of the Verified Projection Engine

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Abstract

Combinatorial maximalism is the methodological stance that when a system's behavioural space is finite by construction — closed vocabularies, enumerated flags, discrete file states — correctness claims must be discharged by *exhaustion over the whole space*, never by sampling, narration, or review. This thesis formalises the stance as practised in the construction of a deterministic graph-to-filesystem projection engine governed by the Chatman Equation $A = \mu(O)$. We give the configuration space of the write discipline an exact product structure and prove total coverage of its 144 cells against an independent reference model; we organise pack composition as a simplicial complex whose $\binom{8}{2} = 28$ edges and single top face were verified collision-free; and we develop the delta calculus in which projection idempotence, replay, and drift acquire the standard shapes of fixed points, integrals, and metrics. Three empirical findings anchor the mathematics: an aggregate that failed to be a *function* of its inputs (the GROUP_CONCAT incident), a fixed point that held at the level of states but not of annotations (the **force** finding), and the predictive success of the exhaustive matrix, whose one flagged tension was independently re-encountered as a live defect before the ink was dry. The central theorem of the thesis is sociotechnical as much as mathematical: under context-bounded construction, *exhaustion is cheaper than trust*, and every place where the space was not exhausted is exactly where the defects lived.

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Chapter 1

The Maximalist Principle

1.1 The stance

Let a system expose a behaviour $b : X \rightarrow Y$ where the input space X is *finite by construction*. The combinatorial maximalist asserts:

Axiom 1.1 (Exhaustion). If $|X| < \infty$ and membership in X is decidable, then the only admissible verification of a specification $\hat{b} : X \rightarrow Y$ is the total check

$$\forall x \in X : \quad b(x) = \hat{b}(x),$$

executed against the running system, with the witness for every x recorded. Sampling, review, and reported success are inadmissible whenever exhaustion is feasible.

Axiom 1.2 (Independent reference). The specification $\hat{b}$ must be computed by an artifact *other than* b — a reference model written from the documented law, not read off the implementation. Verifying b against itself is the self-certification failure mode.

Axiom 1.3 (Closure precedes exhaustion). Exhaustion is only possible over spaces that have been deliberately *closed*: vocabularies that refuse unknown members by name, enumerations with no wildcard arms, flag sets declared rather than discovered. Closure is therefore not a stylistic preference but the precondition of proof.

Axiom 1.3 explains why the same three engineering laws (closed vocabulary; status earned by evidence; chained history) precipitated independently in five sibling systems: they are the minimal conditions under which Axiom 1.1 can be applied at all.

1.2 Why maximalism, not minimalism

The classical testing doctrine is minimalist: choose representative points, partition by equivalence classes, argue the rest by analogy. That doctrine presumes an analyst who holds the whole space in mind while choosing the representatives. The empirical record of the construction examined here shows the presumption false: every defect arose at a *context boundary*, where no single participant — human or model — held both halves of an interaction. When the analyst's memory is the weakest component, the partition argument inherits its weakness. Exhaustion removes the analyst from the trusted base.

Remark 1.4. Maximalism is not profligacy. The spaces exhausted below have cardinalities 144, 28, 10, and 8: closure keeps them small. The maximalist bet is that *designing for small closed spaces and then exhausting them* beats designing for open spaces and then sampling them.

Chapter 2

Algebra: the Structure of the Exhausted Spaces

2.1 The write discipline as a product space

The engine's write stage decides, for each rendered artifact, one of a closed set of outcomes. Its inputs form a product of finite dimensions:

Definition 2.1 (Write configuration space).

$$\Sigma = F \times U \times I \times S \times T$$

with $F = \{\perp, \top\}$ (**force**), $U = \{\perp, \top\}$ (**unless_exists**), $I = \{\perp, \top\}$ (**inject**), $S = \{\emptyset, s=, s\neq\}$ (**skip_if** absent, matching, non-matching), and T the target-file state. For $I = \perp$, $T = \{\text{absent}, \text{identical}, \text{different}, \text{anchored}\}$; for $I = \top$ the anchored state refines into the anchor modes $\{\text{append}, \text{before}, \text{after}, \text{at}, \text{at}_{\text{obj}}\}$. The realised cardinality is

$$|\Sigma| = 2 \cdot 2 \cdot 3 \cdot (4 + 3 + 5) = 144.$$

Definition 2.2 (Decision codomain). $\mathcal{D} = \{\text{Written}, \text{Injected}, \text{Skip}_r, \text{Refuse}_c\}$, where r ranges over named skip reasons and c over the closed error-code set $\{\text{FM-WRITE-002}, \dots, 005\}$.

Definition 2.3 (First-match-wins composition). The decision function is a *priority composition* of guarded rules $\rho_1 \triangleright \rho_2 \triangleright \dots \triangleright \rho_k$:

$$w(\sigma) = \rho_{j(\sigma)}(\sigma), \quad j(\sigma) = \min\{i : \sigma \in \text{dom}(\rho_i)\}.$$

Priority composition is associative and idempotent but *not* commutative; the order of the rule list is semantics.

Theorem 2.4 (Total coverage of the write matrix). *Let $\hat{w}$ be the reference model transcribed from the documented decision table ($\text{escape} \rightarrow \text{FM-WRITE-002}$; **unless_exists** $\rightarrow \text{skip}$; **skip_if** match $\rightarrow \text{skip}$; **inject** family $\rightarrow \text{Injected}$ or FM-WRITE-003/004 ; **force** $\rightarrow \text{Written}$; **absent** $\rightarrow \text{Written}$; **identical** $\rightarrow \text{Skip}$; **different** without **force** $\rightarrow \text{FM-WRITE-005}$). Then, as executed against the real filesystem,*

$$\forall \sigma \in \Sigma : w(\sigma) = \hat{w}(\sigma),$$

with a cardinality guard $\#\{\sigma \text{ exercised}\} = 144$ preventing silent shrinkage of the exhausted space.

Finding 2.5 (Force precedes identity). The single tension surfaced by Theorem 2.4 is not a counterexample but a precedence fact: because $\rho_{\text{force}} \triangleright \rho_{\text{identical}}$ in Definition 2.3, a configuration with $F = \top$ and $T = \text{identical}$ yields **Written**, never **Skip**_{unchanged}. The implementation matches the law; the law diverges from the naïve expectation. Chapter 4 shows this is a distinction between two *grades* of idempotence.

2.2 The vocabulary group and the refusal boundary

Definition 2.6 (Closed vocabulary). The frontmatter vocabulary is a finite set $V = \{\text{to}, \text{sparql}, \text{construct}, \dots\}$ with $|V| = 12$, together with a refusal map $\neg V \rightarrow \text{Refuse}_{\text{FM-TPL-002}}$ that names the offending key.

Exhaustion here has two sides: all 12 members admit; a chosen adversarial sample of 10 plausible non-members (`vars`, `sh_before`, `backup`, `from`, `mode`, ...) refuses by name. The complement of a closed set is infinite, so Axiom 1.1 applies only to V itself; the non-member check is a boundary probe, and honesty requires saying so.

2.3 The delta group

Definition 2.7 (Delta group). For canonical graphs O, O' over vocabulary V , define $\Delta(O, O') = (A, D)$ with $A = O' \setminus O$, $D = O \setminus O'$. Deltas compose by

$$(A_1, D_1) \oplus (A_2, D_2) = ((A_1 \setminus D_2) \cup A_2, (D_1 \setminus A_2) \cup D_2),$$

with identity $\emptyset = (\emptyset, \emptyset)$ and inverse $(A, D)^{-1} = (D, A)$.

Proposition 2.8 (Verified group laws). *As property-tested over randomly generated graph pairs: $\Delta \oplus \Delta^{-1} = \emptyset$; $(\Delta^{-1})^{-1} = \Delta$; $O \oplus \Delta(O, O') \simeq O'$ (replay identity, checked by hash equality).*

2.4 The hash homomorphism and the incident of the non-function

Definition 2.9 (Canonical hash). Let $\approx$ be the congruence generated by blank-node relabelling and insertion order. The state hash is $H = \text{BLAKE3} \circ \text{c14n} : \mathbf{Ont} / \approx \rightarrow \{0, 1\}^{256}$, where `c14n` sorts canonical N-quads (with a bounded colour-refinement pass for blank nodes).

Proposition 2.10 (Well-definedness, exhaustively probed). *H is constant on $\approx$ -classes: verified under (i) permutation of insertion order and (ii) blank-node renaming, both as property tests over generated instances.*

Finding 2.11 (The `GROUP_CONCAT` incident: an aggregate that was not a function). The self-hosting layer projected a CLI route's flag list via `GROUP_CONCAT` over an *unordered* solution multiset. SPARQL guarantees no aggregate ordering; the projection was therefore a *relation*, not a function, on the quotient by solution order. The realised failure was maximally quiet: the two flags were both `bool`, so the swapped signature `sync_run(watch, dry_run)` type-checked against the handler's fixed `(dry_run, watch)` and inverted the semantics of both flags at the CLI. The repair — an inner `ORDER BY` subquery — is precisely the completion of the diagram

$$\text{solutions} \xrightarrow{\text{sort}} \text{sequence} \xrightarrow{\text{concat}} \text{string},$$

restoring functionality by factoring through a canonical total order. Moral, in the algebraic register: *every projection out of a quotient must be shown constant on equivalence classes, and where the standard gives you a multiset, you must pay for the order yourself.*

2.5 Receipt chains as a monoid homomorphism

Definition 2.12 (Chain functional). Histories form the free monoid E^* over receipt payloads. The chain map $R : E^* \rightarrow \{0, 1\}^{256}$ is the left fold $R(\epsilon) = g$ (genesis, the all-zero word) and $R(w \cdot e) = \text{BLAKE3}(R(w) \parallel \phi(e))$, with ϕ the canonical payload serialisation.

Proposition 2.13 (Tamper evidence, adversarially exhausted). *For each index i of a three-link chain, mutation of any field of e_i changes R of every prefix extending i ; the verifier names the least broken index. Verified for payload mutation, chain-hash mutation, truncation, and reordering — the full (small, closed) set of single-step tamper moves.*

Chapter 3

Geometry: Composition Space and Drift

3.1 The pack complex

Definition 3.1 (Pack composition complex). Let $\mathcal{P} = \{p_1, \dots, p_8\}$ be the pack set. For $S \subseteq \mathcal{P}$ let sync_S denote projection of a consumer project admitting exactly S . Say S is *coherent* if sync_S succeeds with all packs' distinctive outputs present and no output-path collision. The coherent subsets form a simplicial complex $K \subseteq 2^{\mathcal{P}}$ (coherence is inherited by subsets, since removing a pack removes only its own contributions).

Theorem 3.2 (Verified skeleton and top face). *The following faces of K were verified by execution: all 8 vertices (single-pack syncs); all $\binom{8}{2} = 28$ edges (pairwise syncs, fresh state each); and the top face $\mathcal{P}$ itself (the mega-project), including lockfile enumeration, receipt coverage of all eight content hashes, and second-sync idempotence.*

Remark 3.3 (Why edges suffice between the vertices and the top). Full exhaustion of $2^8 = 256$ subsets was not performed, and the omission is principled: pack interactions in this engine factor through exactly two channels — ontology union and output-path union — and both are *pairwise-generated*: a collision or prefix clash among k packs implies one among some pair. Hence coherence of all edges plus the top face (which exercises the union of all channels simultaneously) covers the lattice. Where interactions cease to be pairwise-generated — e.g. future **construct** rules of one pack feeding **selects** of another — this argument lapses and the maximalist must widen the exhausted skeleton. The boundary of the method is stated, not hidden.

Proposition 3.4 (Order invariance). *The graph-union channel is a commutative idempotent monoid $(\mathbf{Ont}, \cup, \emptyset)$; consequently declaration order of packs is immaterial, verified by syncing reversed declarations and obtaining a byte-identical receipt payload.*

3.2 Drift as distance; admission as membership

Definition 3.5 (Drift metric). On canonical graphs, $d(O, O') = |O \triangle O'|$, the symmetric difference; equivalently $d(O, O') = |A| + |D|$ for $\Delta(O, O') = (A, D)$. On artifacts, drift is measured per file by hash inequality against the receipt's recorded output map.

The engine's health verbs are radius measurements in three projections of one state space: lock drift (d between locked and on-disk pack content), staleness (d between receipted outputs and disk), and orphanhood (outputs whose preimage under the current template set

is empty — a point outside the image of μ). Refusal is the assertion $d > 0 \Rightarrow$ named error; the corruption sweep verified that perturbing any single pack moves exactly one coordinate and that the refusal names it and nothing else.

Definition 3.6 (Feasible region). Admission constraints (SHACL shapes, **when** guards, closed vocabularies) carve a feasible region $\mathcal{F} \subseteq \mathbf{Ont}$; μ is a partial map with $\text{dom}(\mu) = \mathcal{F}$, and fail-closed means the complement of $\mathcal{F}$ maps to named refusals rather than to best-effort artifacts.

3.3 Projection as retraction

Theorem 3.7 (Idempotence as fixed point). *For fixed admitted inputs (O^*, C^*, P^*, T^*) , the map $F \mapsto \text{sync}(F)$ is a retraction of filesystem space onto its image: $\text{sync} \circ \text{sync} = \text{sync}$, verified at byte level (output files identical) and at payload level (graph hash, output hashes, pack hashes identical) across repeated runs and across independently created state (property-tested over generated projects).*

Geometrically: the admitted state $F^* = \text{sync}(F^*)$ is the unique fixed point in each fibre, and every sync is a projection onto it. Drift moves you off the fibre; the next sync either brings you back (**force**) or refuses by name (default) — but never silently splits the difference.

Chapter 4

Calculus: Deltas, Replay, and the Two Grades of Idempotence

4.1 Differentiation and integration of history

Definition 4.1 (Discrete derivative and integral). For an ontology trajectory $O_0, O_1, \dots, O_n$, the derivative is $\Delta_i = \Delta(O_{i-1}, O_i)$ and the integral is replay:

$$O_n = O_0 \oplus \bigoplus_{i=1}^n \Delta_i,$$

a discrete fundamental theorem verified term-by-term against the receipt chain: R of the prefix at step i must equal the prev-pointer of step $i + 1$, genesis-rooted.

Proposition 4.2 (Chain rule across the pipeline). *Writing $\mu = \mu_5 \circ \dots \circ \mu_1$ (resolve, admit, enrich, extract, render/write), a perturbation ΔO propagates as $\Delta A \subseteq \mu'_5 \cdot \mu'_4 \cdots \mu'_1 (\Delta O)$: each stage’s sensitivity bounds which artifacts a graph change may touch. The enrichment stage μ_3 is a closure operator (extensive, monotone, idempotent), so its “derivative” at a fixed point is the identity on already-entailed triples — the formal content of “running enrichment twice adds nothing.”*

4.2 The two grades of idempotence

Finding 2.5 and its live recurrence (a freshly authored pack shipped with `force: true` and failed the second-sync gate) force a distinction the naïve statement of Theorem 3.7 blurs.

Definition 4.3 (State- and trace-idempotence). Let `sync` act on states F and also emit an annotation trace $\tau(F) \in \mathcal{D}^*$ (the per-file decisions bound into the receipt). Call the system *state-idempotent* if $\text{sync}(\text{sync}(F)) = \text{sync}(F)$, and *trace-idempotent* if additionally $\tau(\text{sync}(F)) = \tau(\text{sync}(\text{sync}(F)))$ up to the intended $\text{Written} \rightarrow \text{Skip}_{\text{unchanged}}$ transition.

Finding 4.4 (Force is state-idempotent but not trace-quiescent). With `force: true` the second sync rewrites identical bytes: the state fixed point holds, the byte comparison holds, but the trace reports `Written` forever and the file’s modification time advances each run. Under a filesystem watcher this non-quiescence is not cosmetic: it is a self-excitation term, and the engine’s watch mode oscillated on exactly this coupling until the lockfile write was made trace-quiescent (skip when bytes match). The calculus lesson: *a fixed point of the state map is insufficient; the closed loop formed with any observer of side-channels (mtimes, events, logs) requires quiescence of the trace as well*. Formally, the watch loop computes $F_{k+1} = \text{sync}(F_k)$ triggered by $\text{events}(F_k \rightarrow F_{k+1})$; convergence demands $\text{events}(F^* \rightarrow F^*) = \emptyset$, a strictly stronger condition than F^* being fixed.

4.3 Sensitivity of the guard cross

The $2 \times 2 \times 2$ cross of **when**-guard truth, result-set emptiness, and **skip_empty** was exhausted by execution (eight cells, each asserting both the decision string and the presence/absence of the file). In calculus terms this is the verification that the piecewise definition of μ has no unglued seams: on each cell the projection restricts to the documented branch, and the branches agree on their shared boundaries (empty result with guard false is skipped for the guard's reason, not the emptiness's — precedence again, exhaustively pinned).

Chapter 5

The Maximalist Theorems

Theorem 5.1 (Exhaustibility). *If a subsystem’s inputs are confined to a product of closed dimensions (Axiom 1.3), then total verification against an independent reference model (Axiom 1.2) is decidable in $O(|\Sigma|)$ executions, and $|\Sigma|$ is a design variable: closure decisions made for refusal’s sake simultaneously make proof cheap.*

Theorem 5.2 (Predictive power of exhaustion). *An exhaustive matrix does not merely confirm; it forecasts. The one tension it flagged (Finding 2.5) was re-encountered as a live defect in independently authored code before any consumer existed. A sampled suite that skipped the $F = \top$, $T = \text{identical cell}$ — the natural cell to skip, since “force obviously overwrites” — would have shipped the oscillation.*

Theorem 5.3 (Locality of the defect set). *Every defect in the construction record lay outside some exhausted space at the moment of its introduction: the aggregate-order relation (no ordering dimension had been closed), the manifest clobber (no exhausted check compared disk to intent across sessions), the lockfile self-excitation (the trace dimension was not yet in the matrix), the pack **force** recurrence (pack authors’ space excluded the write-precedence dimension). Conversely, no defect was found inside a space after its exhaustion. The maximalist reading: the boundary of the exhausted region is the support of the defect distribution.*

Corollary 5.4 (The programme). *Verification effort should be spent enlarging the closed, exhausted region — adding dimensions (trace quiescence, aggregate order, cross-session disk state) to Σ — rather than deepening scrutiny inside it. Each of the 2028–2030 frontier items (triple admission gating, receipt federation) is the closure of one presently-open dimension.*

Chapter 6

Conclusion

Combinatorial maximalism is often mistaken for brute force. This thesis has argued it is the opposite: a *design* discipline in which vocabularies are closed so that spaces become finite, spaces are given product structure so that exhaustion is enumerable, and exhaustion is executed against independent reference models so that no participant’s memory — human or machine — remains in the trusted base. The algebra supplied the objects (priority compositions, delta groups, hash homomorphisms on quotients, chain monoids); the geometry supplied the shapes (the pack complex, the drift metric, projection as retraction); the calculus supplied the dynamics (replay as integration, the chain rule of the pipeline, and the two grades of idempotence whose gap is precisely where a live system oscillates).

The empirical record closes the argument. Where the space was exhausted, no defect survived. Where a dimension was still open, a defect was waiting — quiet, type-correct, and confidently reported as done. Between those two observations lies the whole thesis:

The boundary of what you have exhausted is the frontier of what you may trust.

Appendix A

Empirical Ledger

Exhausted space	Structure	Cardinality
Write-decision matrix	product $F \times U \times I \times S \times T$	144
Pack pairwise skeleton	edges of K on 8 vertices	28
Pack vertices + top face	$K^{(0)} \cup \{\mathcal{P}\}$	$8 + 1$
Guard cross	when $\times$ emptiness $\times$ skip_empty	8
Vocabulary admission	members of V	12
Vocabulary boundary probe	adversarial non-members	10
Tamper moves on a 3-chain	field/chain/truncate/reorder per index	all

Property-tested laws (generated instances): state-hash invariance under insertion order and blank-node renaming; delta cancellation $\Delta \oplus \Delta^{-1} = \emptyset$ and involution; replay identity; byte- and payload-level sync idempotence. Engine: `praxis/crates/ggen` v26.7.4, 182 tests, zero test doubles; commits `f011738`, `71699f8`, `314db73`. Formal substrate: `docs/ggen-theory.md`; companion prospectus: `docs/chatman-equation-phd-thesis.md`.